

RESEARCH ARTICLE

Performance of XGBoost Ensemble Learning Algorithm for Mangrove Species Classification with Multisource Spaceborne Remote Sensing Data

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Accurately and timely monitoring the spatial distribution and composition of mangrove species is critical for assessing mangroves' health, dynamics, and biodiversity, as well as mangroves' degradation and restoration. Recent advances in machine learning algorithms, coupled with spaceborne remote sensing technique, offer an unprecedented opportunity to map mangroves at species level with high resolution over large extents. However, a single data source or data type is insufficient to capture the complex features of mangrove species and cannot satisfy the need for fine species classification. Moreover, identifying and selecting effective features derived from integrated multisource data are essential for integrating high-dimensional features for mangrove species discrimination. In this study, we developed a novel framework for mangrove species classification using spectral, texture, and polarization information derived from 3-source spaceborne imagery: WorldView-2 (WV-2), Orbital HyperSpectral (OHS), and Advanced Land Observing Satellite-2 (ALOS-2). A total of 151 remote sensing features were first extracted, and 18 schemes were designed. Then, a wrapper method by combining extreme gradient boosting with recursive feature elimination (XGBoost-RFE) was conducted to select the sensitive variables and determine the optimal subset size of all features. Finally, an ensemble learning algorithm of XGBoost was applied to classify 6 mangrove species in the Zhanjiang Mangrove National Nature Reserve, China. Our results showed that combining multispectral, hyperspectral, and L-band synthetic aperture radar features yielded the best mangrove species classification results, with an overall accuracy of 94.02%, a quantity disagreement of 4.44%, and an allocation disagreement of 1.54%. In addition, this study demonstrated important application potential of the XGBoost classifier. The proposed framework could provide fine-scale data and conduce to mangroves' conservation and restoration.

Introduction

Mangroves are a specific wetland community that occurs in the tropical and subtropical coastal regions of 118 countries worldwide. They provide essential resources, environmental and ecological benefits that are crucial to biodiversity conservation, climate change mitigation, and human well-being maintenance [1]. Mangrove forests are also one of the most vulnerable and threatened ecosystems [2,3]. Although countries across the globe have made extensive efforts to protect and restore of mangroves, large-scale mangrove degradation remains an ongoing

trend under the impacts of climate change and human activities [4,5]. Mangrove species information provides valuable insights into the health and evolution of mangrove ecosystems and is also a sensitive indicator of both mangrove degradation and restoration [6]. Monitoring the spatial distribution and composition of mangrove species is crucial for sustainable development of mangrove forests [7]. Therefore, rapid and precise technique for mapping mangrove species is highly desirable.

The remote sensing technique is an effective substitute for traditional field surveys and has been widely applied for mangrove species classification [7–9]. However, because of the high

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spatial heterogeneity and complexity among mangrove species, high-precision and large-scale mapping of mangrove species with remote sensing technique remains a challenging task [10]. Both remote sensing data and classification algorithm are key factors that determine the accuracy of mangrove species classification [6,7]. The most popular data source for mangroves species classification is optical high-resolution remote sensing data, such as satellite-based data from WorldView (WV) series [11]. Spectral reflectance features (SRFs) derived from WV-2 imagery were used to classify 24 mangrove species in Karimunjawa and Kemujan Island of Indonesia [12]. However, in most cases, the ability of multispectral image is limited, resulting in unsatisfactory classification results with the highest overall accuracy (OA) of 49.72% [12]. Furthermore, the strong spectral similarity between mangrove communities makes it difficult to distinguish different community types using optical remote sensing data alone. Hyperspectral remote sensing images consist of hundreds of continuous narrow bands with rich spectral information, which have been proven to be helpful for identifying mangroves with high species diversity [13,14]. Commonly, existing researches have mostly utilized unmanned aerial vehicle (UAV)-based hyperspectral data, which are generally limited by high costs and small areas of data obtaining; in contrast, spaceborne hyperspectral imagery may be limited in species-specific identification owing to their spatial heterogeneity.

Moreover, cloud contamination has a substantial impact on optical remote sensing images and can lead to data loss, particularly in areas with large-scale and cloudy/rainy mangroves. As an active technique to acquire remote sensing data, synthetic aperture radar (SAR) can obtain data under all-weather conditions and provide structural information (e.g., density and morphology) of vegetation [15]. SAR images commonly used for vegetation studies include C-band, L-band, and P-band, with different bands exhibiting varying degrees of sensitivity to vegetation information. Numerous experiments have shown that C-band or X-band radar is suitable for herbaceous vegetation, whereas longer wavelength L-band radar is better suited for forests [16]. The scattering characteristics of mangrove species were quantified on the basis of dual-polarization Sentinel-1 images, showing that SAR images and their derivatives improved the ability to identify different mangrove species [17]. Combining the Advanced Land Observing Satellite phased array type L-band SAR (ALOS PALSAR) images and Geographic Information System (GIS) datasets has been used to map mangrove species along the coast of Hai Phong [18]. Nonetheless, these studies mainly used C-band SAR data or only used single features of L-band SAR data.

In fact, a single data source or data type is insufficient to capture the complex features of mangrove species and cannot satisfy the need for fine species classification. Hence, many researchers have investigated the potential of multisource remote sensing data to improve the classification accuracy of mangrove species, focusing on the integration of data from UAV-based platforms, such as UAV-multispectral, UAV-hyperspectral, and UAV-light detection and ranging sensors [7,9,13,19]. In addition, many UAVs are limited in their ability to obtain images in the visible and near-infrared (NIR) spectrum. To our knowledge, little attention has been paid to integrating multisource spaceborne-based remote sensing data of multispectral, hyperspectral, and SAR data for mangrove species classification.

With the development of remote sensing data and recent advances in machine learning (ML) technology, more and more

ML algorithms have been introduced in mangrove remote sensing research [20]. Compared with traditional methods, ML is proficient in handling nonlinear problem, has higher model accuracy and computational efficiency [9], and is more suitable for analyzing complex ecosystems like mangroves. Random forest (RF), support vector machine (SVM), artificial neural network, etc. are currently successful and widely applied in mangroves mapping [14]. In particular, ensemble learning algorithms like RF, which combine the results of multiple base classifiers to achieve better predictive performance [21], have emerged as an influential research area in the field of ML. This is due to their strong ability to overcome the limitations of a single classifier. Besides, numerous derived features can be generated from such multisource data, leading to data redundancy and reduced efficiency [6]. Identifying and selecting effective features for mangrove species discrimination, therefore, are essential for integrating high-dimensional features. More recently, some studies have achieved promising results by utilizing ML algorithms that combine feature selection and classification capabilities [6,22]. Extreme gradient boosting (XGBoost) is a state-of-the-art ML algorithm that exhibits exceptional performance owing to its scalability in all situations and its ability to rapidly process large datasets [23]. However, only a few scholars have used the XGBoost algorithm for fine-scale classification of mangrove species, and there is a lack of quantitative assessment of its effectiveness and reliability using multisource remote sensing data.

Therefore, this study took the Zhanjiang Mangrove National Nature Reserve (ZMNNR), China, as the study area and developed a framework for mapping mangrove species. The specific objectives are (a) identifying sensitive features of WV-2, Orbital HyperSpectral (OHS), and ALOS-2 spaceborne images to classify mangrove species using XGBoost with recursive feature elimination (XGBoost-RFE) algorithm; (b) evaluating the performance of the XGBoost algorithm to construct pixel-based classification model under high-dimensional features derived from multisource remote sensing data; and (c) analyzing the contributions of multisource remote sensing data to identify mangrove species. The framework proposed in this study will contribute to the identification of mangrove species information at a large scale.

Materials and Methods

Study site and ground-truth data collection

The study site (21°31' to 21°35'N, 109°45' to 109°48'E) is one of the core zones of the ZMNNR, China (Fig. 1). Details regarding the climatic conditions in the study area can be found in our previous study [24]. The mangroves in the study area exhibited high coverage and obvious zonal characteristics, consisting of 6 dominant true mangrove species: *Aegiceras corniculatum* (AC), *Bruguiera gymnorhiza* (BG), *Kandelia obvolata* (KO), *Rhizophora stylosa* (RS), *Avicennia marina* (AM), and *Sonneratia apetala* (SA). These species exhibit distinct tree morphologies and leaf greenness (Fig. 1). They are mostly natural secondary and artificial forests, with only a few parts of the area having primary forests. We conducted 5 field surveys during May 2018, January 2019, August 2019, December 2019, and August 2020. Altogether, 599 sampling points were collected, of which 68 were AM, 192 were BG, 65 were RS, 229 were AC, 29 were SA, and 16 were KO. Because of limited human accessibility for fieldwork, additional 7,986 samples were collected by visual interpretation

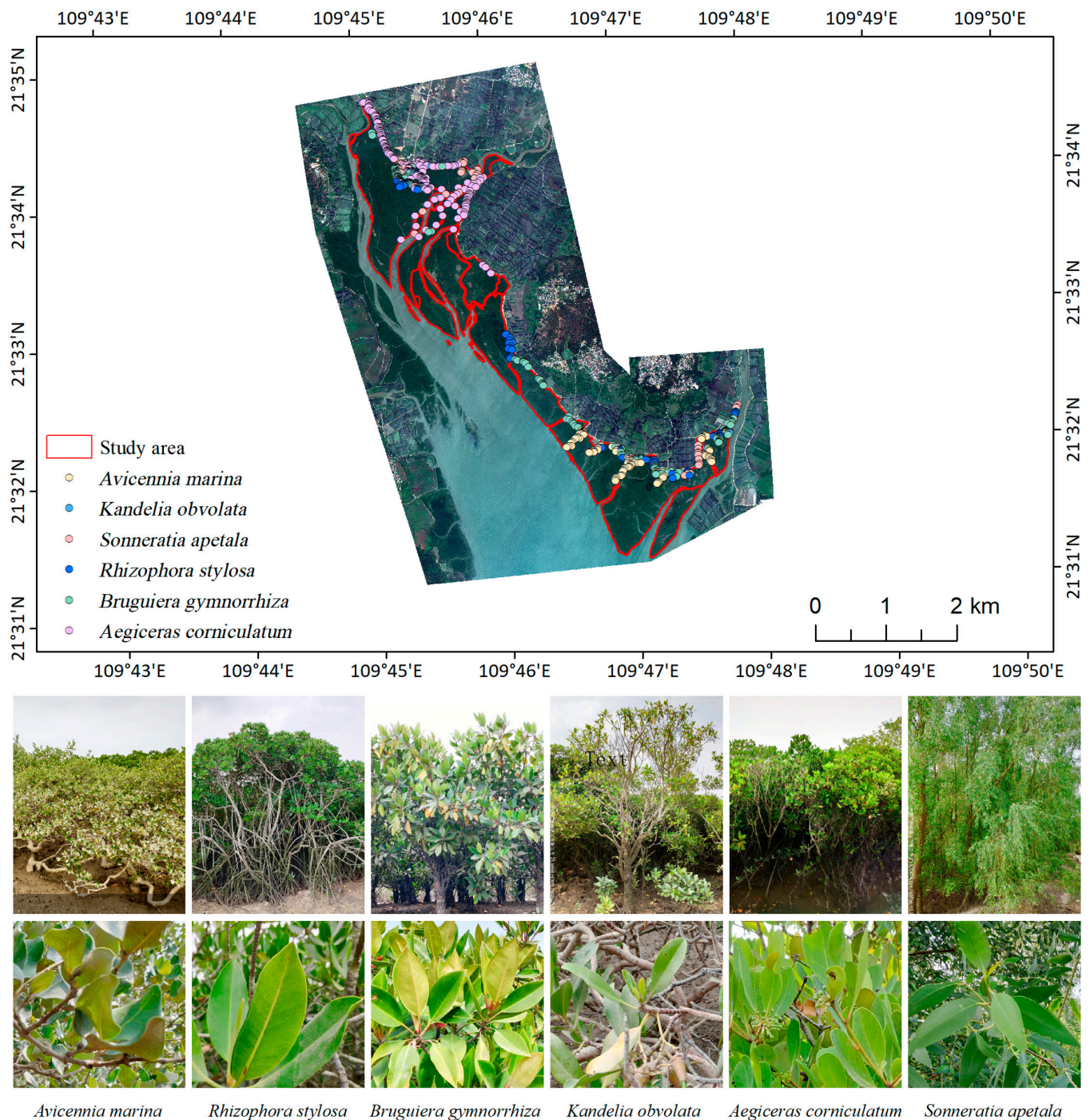


Fig. 1. Location of study area.

of Google Earth and UAV images. The UAV image was obtained on January 2019 by a DJI Phantom 4 multispectral drone that uses an integrated multispectral imaging system. The flying altitude was 180 m above ground level, obtaining the hyperspectral image with a spatial resolution of 0.098 m.

Satellite data acquisition and preprocessing

WV-2 data

A level 2A WV-2 image was collected on 7 October 2020. The WV-2 sensor records 8 multispectral bands with a spatial

resolution of 2 m and a panchromatic band with a spatial resolution of 0.5 m (Table S1). Radiometric correction was performed first to convert the digital number (DN) values to the top-of-atmosphere spectral radiance, and atmospheric correction was followed by the Fast Line-of-Sight Atmospheric Analysis of Hypercubes correction method to obtain the surface reflectance. Orthorectification was then conducted using the Rational Polynomial Coefficients (RPC) information contained in the image and the Digital Elevation Model (DEM) data. To link the field survey samples and training samples with image for mangrove species classification, the multispectral images were fused

with the panchromatic image via the NNDiffuse Pan Sharpening method in ENVI 5.3 to generate a 0.5-m pansharpened image.

OHS data

OHS is the second group satellites of the Zhuhai-1 satellite constellation operated by a Chinese company (Zhuhai Obita Aerospace Technology Co. Ltd.). It consists of 32 spectral channels covering a spectral range of 400 to 1,000 nm with a spectral interval of 3 to 8 nm (Table S2). An level 1A 10-m OHS image was acquired on 19 July 2020 at a similar time when the WV-2 image was collected, which was processed with systematically geometrical calibration and provided an RPC file. Similar to WV-2, radiometric correction, atmospheric correction, and orthorectification preprocessing steps were applied to OHS image. The surface reflectance of the OHS image was then resampled to 2 m and registered to the WV-2 image, with an error of less than one pixel.

ALOS-2 data

A level 1.1 dual-polarization [horizontal transmit and horizontal receive (HH) and horizontal transmit and vertical receive (HV)] radar data of the ALOS-2 PALSAR was collected on 18 October 2018. The image was captured in the single-look complex format at an incidence angle of 40.552°, with a range spacing of 4.29 m and an azimuth spacing of 3.42 m. To square the pixels in the ground range geometry and reduce the speckle noise, the ALOS-2 data were first multilooked [25], with 2-look in the azimuthal direction and 1-look in the range direction, to obtain the target pixel size of 6.7 m. The DN image was then calibrated to backscatter sigma0 (σ^0) using Eq. 1 [26].

$$\sigma_{1.1 \text{ product}}^0 (\text{dB}) = 10 \cdot \log_{10}(\text{DN})^2 + \text{CF} - A \quad (1)$$

where σ^0 is sigma0 backscattering intensity and DN is the digital number value. For complex SAR data, $\text{DN} = I^2 + Q^2$, where I and Q are the real and imaginary parts of the single-look complex product, respectively. CF is the calibration factor (−83.0 dB for ALOS-2 data); $A = 32.0$ is the conversion factor.

Thereafter, the image was terrain-corrected and geocoded to the WGS84_UTM_zone_49N map projection, which was the same map projection with WV-2 and OHS images. To further suppress speckle noise, a refined Lee filter with a 7×7 window size [25] was applied to the ALOS-2 data. As such, the ALOS-2 image was resampled to 2 and 10 m and then coregistered to the WV-2 and OHS images, respectively, with an error of less than one pixel.

Auxiliary data

The 2-m DEM data were applied in this study as auxiliary information, which was generated by the 3-dimensional analysis tool in ArcGIS10.4 software processing the elevation point data downloaded from LocaSpaceViewer (<http://www.tuxingis.com>).

Mangrove species classification methodology

The classification procedure consists of 3 steps: (a) feature generation, (b) feature selection, and (c) modeling and validation. The comprehensive framework is presented in Fig. 2.

Feature generation

In this study, the feature variables used for mangrove species classification were divided into 5 categories: (a) SRFs, (b) vegetation indices (VIS), (c) texture features (TFs) for optical images (WV-2 and OHS), (d) backscatter intensity

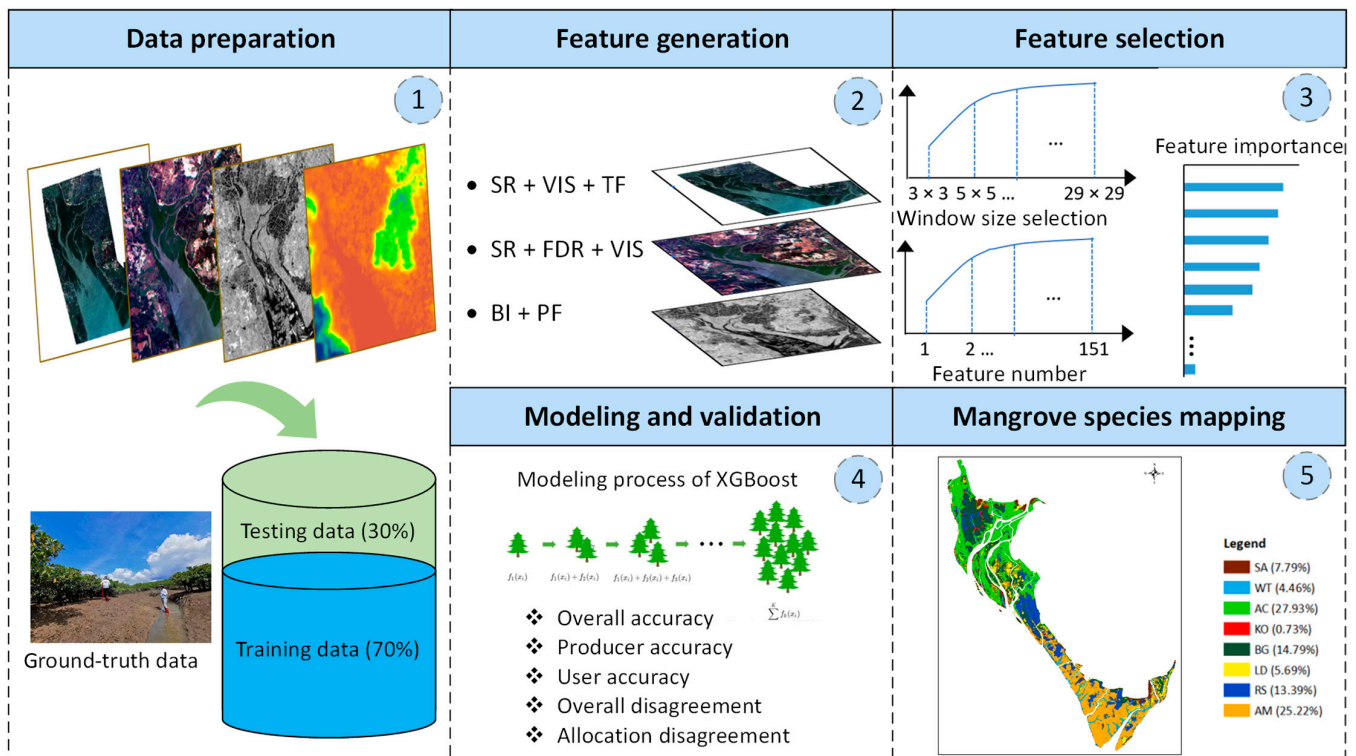


Fig. 2. Flowchart of the methodology implemented in this study.

(BI), and (e) polarimetric features (PFs) for radar image (ALOS-2).

SRFs comprised 40 original spectral reflectance (SR) bands (8 from WV-2 and 32 from OHS) and 32 first derivative reflectance (FDR) bands of OHS, which contributed to enhancing the differences between the original spectral bands and eliminated background noise, calculated by Image Derivative Tool in ENVI5.3. We also derived 16 VIS from the WV-2 image and 22 from the OHS images (Table S3).

The Gray Level Cooccurrence Matrix (GLCM) is a classic method that extracts texture by calculating the conditional probability density between image gray levels [27]. Texture measures, the size of the analysis window, and the basis image are 3 crucial issues when GLCM analysis is performed [28]. In this study, textural metrics were generated from GLCM based on the first band of principal components analysis [29] of the original WV-2 image. A total of 28 texture measures, including 8 simple, 10 advanced, and 10 higher-order features (see Table S4), were extracted using the Haralick Texture Extraction tool in an open-source remote sensing software OTB (Orfeo ToolBox). We conducted 100 randomized experiments applying 15 different window sizes (ranging from 3×3 to 29×29) to explore the optimal texture measures and the associated window size for producing the texture image. All TFs were used with 45° orientation, one-pixel statistical interval, and 64-bin histogram number.

To make the most of the ALOS-2 data, HH and HV polarization and their combinations, including HH/HV, HH-HV, and $(HH + HV)/2$ [30], were used in this study. In addition, we calculated polarimetric parameters derived from the $H/A/\alpha$ polarimetric target decomposition technique that has shown important potential for mangroves' species classification [31]. The derived variables also incorporated HA , $H(1 - A)$, $(1 - H)A$, and $(1 - H)(1 - A)$. The effective polarimetric decomposition components [α , entropy (H), and anisotropy (A)] that provide important information based on the scattering mechanism of each SAR image pixel [32] are formulated in Eqs. 2 to 4.

$$\alpha = \alpha(P_1 - P_2) + P_2 \frac{\pi}{2} \quad (2)$$

$$H = P_1 \log_2 P_1 + P_2 \log_2 P_2 \quad (3)$$

$$A = \frac{\lambda_1 - \lambda_2}{\lambda_1 + \lambda_2} \quad (4)$$

where λ_i represents the 3 eigenvalues of the coherence matrix and P_i is calculated by λ_i , namely, $P_i = \frac{\lambda_i}{\lambda_1 + \lambda_2 + \lambda_3}$

Comparison experiments were designed to assess the performance of the different data source and features. The detailed experimental design is listed in Table 1.

Feature selection and modeling

(1) Extreme gradient boosting. The XGBoost algorithm was proposed by Chen and Guestrin [23] with the aim of enabling the boosting tree to break through its own computational limits to achieve the goal of fast computing and state-of-the-art performance. Compared with traditional gradient boosting algorithms, XGBoost has made many improvements and has been considered an advanced evaluator with ultrahigh performance in both classification and regression tasks. The prediction result of each sample in XGBoost is the sum of the leaf weights for all the weak classifiers (as shown in Eq. 5). Another core difference is that XGBoost uses a regularization term to prevent the shortcoming for tree model's inherent tendency to overfitting, allowing the model to be minimally overfitted before pruning.

$$\hat{y}_i = \sum_{k=1}^K f_k(x_i), f_k \in E \quad (5)$$

where $E = \{f(x) = \omega_{q(x)}\} (q: R^m \rightarrow N, \omega \in R^N)$ is the space of regression tree; q is the structure to each classification and regression tree (CART) that maps the example to its corresponding leaf index; and N is the number of leaves in the tree. Here, it is worth noting that ω_i represents the score on each leaf.

Moreover, XGBoost strikes a balance between model performance and computing speed, which is achieved by an objective function represented as a traditional loss function plus a regularization term for controlling model complexity (Eq. 6).

Table 1. Experiment designed in our study

Single-source data						Combined data	
WV-2		OHS		ALOS-2			
Scheme	Input features	Scheme	Input features	Scheme	Input features	Scheme	Input features
1	SR	8	SR	12	BI	15	WV-2 + OHS
2	VIS	9	FDR	13	PFs	16	WV-2 + ALOS-2
3	TFs	10	VIS	14	BI + PFs	17	OHS + ALOS-2
4	SR + VIS	11	SR + FDR + VIS			18	WV-2 + OHS + ALOS-2
5	SR + TFs						
6	VIS + TFs						
7	SR + VIS + TFs						

The first term represents the traditional loss function, which measures the difference between the real value y_i and predicted value $\hat{y}_i$. The second term represents the complexity of the model, expressed using a transformation (Ω) of the tree model, which represents an equation that measures the complexity of the tree model in terms of the tree.

$$\text{Obj} = \sum_{i=1}^m l(y_i, \hat{y}_i) + \sum_{k=1}^K \Omega(f_k) \quad (6)$$

where i represents the i th sample in the dataset, m represents the total amount of data imported into the k th tree, and k represents all trees created.

(2) XGBoost-RFE. The wrapper method is a feature selector that convolves in a classifier (e.g., RF and SVM classifier) and uses search strategies such as genetic algorithm (GA) and RFE for its purpose. In this study, RFE was introduced into the XGBoost algorithm (XGBoost-RFE), which was applied to select important predictors and determine the optimal subset sizes for multisource data. The XGBoost algorithm was used to obtain and rank the importance of the features; then, the features with the lowest importance were removed according to the backward iteration principle. In the XGBoost-RFE algorithm, one feature was removed in each iteration, and the feature selection was stopped when the overall classification accuracy was reduced by less than 0.1% compared with the last accuracy. To enhance model stability and increase the confidence of the feature selection results, we conducted 100 random experiments to determine the optimal features for each scheme.

(3) Optimization of model parameters. Python and the packages “xgboost” were used for model development in this study. The XGBoost model is complicated with more than 20 parameters, among which the key parameters include (a) num_of_boost_round, (b) eta, (c) max_depth, (d) min_child_weight, (e) gamma, (f) subsample, and (g) colsample_bytree. The details of these and other parameters can be found at <https://xgboost.readthedocs.io/en/latest/parameter.html>. Given the large number of parameters in the XGBoost model and the wide range of desirable parameters, the speed of operation and results of tuning the parameters using the GridSearch method may not be promising. Therefore, we use the learning curve combined with the GridSearch method for model tuning to determine the optimal combination of hyperparameters. To ensure sufficient data in each fold and reliable evaluation, we utilized 5-fold cross-validation technique and “mlogloss” as the evaluation metric for hyperparameter combinations in each model construction.

Accuracy assessment

To minimize errors caused by the random division of the training and test sample datasets and ensure that the training and testing datasets had consistent sample classes, we conducted 100 randomized replicate experiments in which we selected the dataset by species using random stratified sampling with 70% for model training and 30% for model validation. In addition to OA, producer accuracy (PA), and user accuracy (UA), which are the most commonly used indicators for remote sensing classification accuracy assessment, we adopted overall disagreement instead of the traditional kappa coefficient. It is the complement of OA and can be decomposed into 2 indicators, allocation disagreement (AD) and quantity disagreement (QD)

(Eqs. 7 and 8); and the smaller the values of AD and QD, the better is the classification model performance. When the value of AD or QD is greater than 10%, the inconsistency it measures is considered significant [33].

$$\text{AD} = \frac{\sum \left(2 \times \min \left(\frac{A_{i+}}{n} - \frac{A_{ii}}{n}, \frac{A_{+i}}{n} - \frac{A_{ii}}{n} \right) \right)}{2} \times 100\% \quad (7)$$

$$\text{QD} = \frac{\sum \left| \frac{A_{i+}}{n} - \frac{A_{+i}}{n} \right|}{2} \times 100\% \quad (8)$$

where A is the confusion matrix, n is the number of mangrove species, i represents the row number of the confusion matrix, A_{i+} represents the total number of samples for species i in that row, and A_{+i} represents the total number of samples for species i in that column.

Results

Spectral and backscatter characteristic of different species of mangrove forests

The 6 mangrove species, AM, RS, BG, AC, SA, and KO, demonstrated typical spectral response characteristics of green plants in both WV-2 and OHS images (Fig. 3A and B). SR in the red edge increased rapidly because of the strong reflectance of the cell tissue inside the leaves to the NIR band. Caused by the changes in the internal structure of leaves and canopy, the near infrared band at 780 to 950 nm showed high reflectance. On the whole, the differences in SR were most evident in the peak (550 nm), red-edge (670–760 nm), and NIR bands (760 to 950 nm). Compared with WV-2, more detailed spectral information is shown in the OHS hyperspectral image.

For all mangrove species (Fig. 3C and D), the median σ^0 values were approximately 5 dB higher in the HH-polarized band than in the HV-polarized band. AM showed the highest

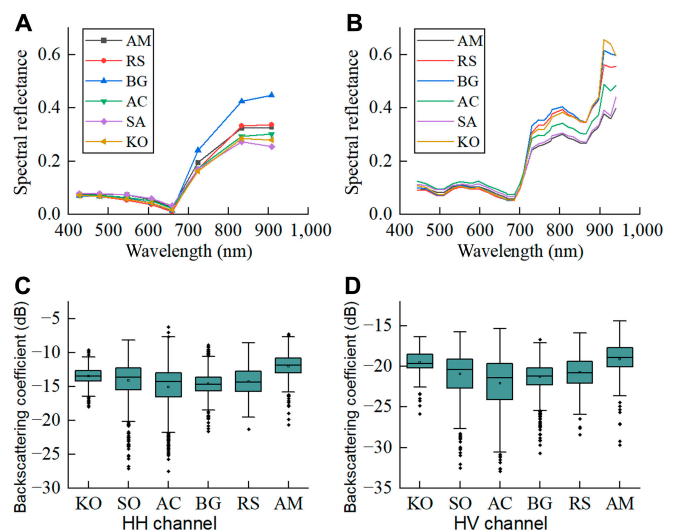


Fig. 3. Spectral and backscatter characteristic of the species mangroves. (A and B) Mean SR in WV-2 and OHS images. (C and D) Box plots of backscattering coefficient at HH and HV channels in ALOS-2 image. The black bars within boxes illustrate the median.

backscattered intensities about -12.01 dB in HH and -19.09 dB in HV. KO and RS also showed high backscattered values but slightly lower than AM, possibly because of similar scattering mechanisms from the upper part of the canopy and volume scattering within them. While lowest values were observed in dense homogenous AC stand -15.11 and -22.05 dB for HH and HV polarizations, respectively. Slightly higher σ^0 values were observed for the SA and BG stands.

Feature selection

Optical window size of TFs on the 2-m WV-2 image

Figure 4 shows the variation in the mean prediction accuracy with the increasing window size. Initially, the accuracy improved rapidly as the window size increased, until the accuracy improvement slowed down at a window size of 15×15 . Our study shows that a customized window size could increase the ability to classify mangrove species. On the basis of our experimental results and the experience of previous research, a window size of 19×19 was selected as the optimal window size for mangrove species classification.

Feature set selection for 18 schemes

For single-source WV-2 data, schemes 1 to 3 show the selected feature variables utilizing only the SR, VIS, or TFs. As shown in Figs. S1 and S2, almost all multispectral bands of WV-2 were preferred, with the order $B6 > B7 > B8 > B4 > B3 > B5 > B2$, among which the red-edge and 2 NIR bands were the top 3 bands. Six indices (MCARI, GNDVI, RVI, DVI, CMRI, and MSAVI) were selected (Figs. S3 and S4). Of the 18 dominant features selected for TFs, there were 6 simple, 6 higher, and 6 advanced TFs (Fig. S6). Schemes 4 to 7 in Table 2 show the feature selection results using combined information from the SR, VIS, and TFs of the WV-2 data. Comparing the 4 integrated feature sets, TFs showed apparent advantages over the others in terms of the number and importance of features. According to the feature ranking in Table 2 and Fig. S14, satisfactory classification results can be obtained by using 22 feature variables among which TFs accounts for 64% and SR and VIS account for 18%.

For single-source OHS data, the optimal predictors for modeling were derived from schemes 8 to 11 in Table 2. It can be seen that among the selected spectral bands and FDR bands (Figs. S15 and S17), approximately 74% of the total number of bands was located in red, red-edge, and NIR regions, which was similar to WV-2 data. In the VIS, the selected indices mainly were calculated by red, red-edge, and NIR region bands and were sensitive to chlorophyll content (MCARI1, MCARI2, and CIg2) and soil background (MSAVI). The results of feature selection in scheme 11 suggested that FDR was the most important, followed by spectral bands and then VIS.

For single-source ALOS-2 data, schemes 12 to 14 in Table 2 show that the top 2 important variables were HV and HH backscatter coefficients, followed by entropy and alpha polarization decomposition variables. Three combinations of polarization features, HH + HV, HA, and $H(1 - A)$, were also among the selected features.

When comparing the contributions of the 3 data sources to mangrove species classification, whether through a combination of 2 or 3 data sources, we found that the TFs of WV-2, FDR of OHS, and several polarization features were the prominent features for accurate classification of mangrove species. As shown in scheme 18 in Table 2 and Fig. S36, of the 151 feature variables extracted from 3 data sources, a total of 62 variables were selected for use in the final model. The WV-2 data contributed more useful features than the OHS and ALOS-2 images. Among these 62 variables, 36 were from WV-2 data, including 6 SR, 6 VIS, and 24 TFs; 22 variables from OHS data, including 8 spectral bands, 1 VIS and 13 FDR bands; and 4 variables from ALOS-2 data, consisting of HH, HH + HV, alpha, and entropy (Fig. S36).

On the basis of the feature selection results, the optimized values for the tuned parameters of each model are listed in Table S5.

Results of mangrove species classification

As shown in Fig. 5, among the 18 schemes, the classification accuracies also varied significantly because of the differences in the data source and input features used. When single-sensor-based

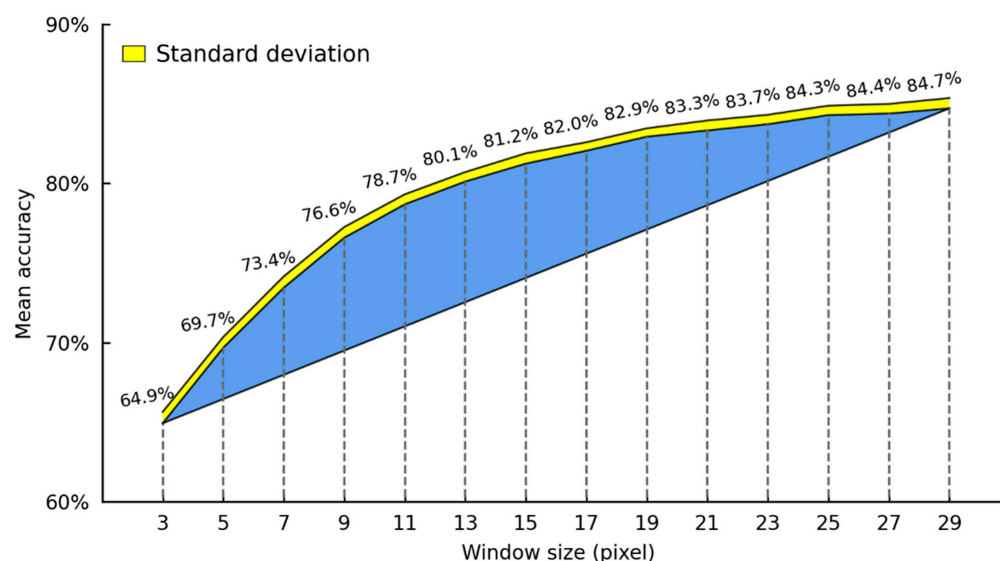


Fig. 4. Relationship between texture window size and mean accuracy for first band of principal components analysis of WV-2 image.

Table 2. Feature selection results of the eighteen classification schemes using WV-2, OHS, ALOS-2 data, and their combined data

No.	Scheme	Resolution (m)	Raw feature size	Optimal feature size	Selected features (listing in descending order of importance)
Single-source data					
1	WV-2 SR	2 m	8	7	B6, B7, B8, B4, B3, B5, B2
2	WV-2 VIS	2 m	16	6	MCARI, GNDVI, RVI, DVI, CMRI, MSAVI
3	WV-2 TFs	2 m	28	18	Simple6, higher3, simple3, advanced5, simple5, advanced9, advanced6, simple4, higher8, higher10, advanced10, higher7, advanced2, advanced1, simple1, higher5, simple2, high6
4	WV-2 SR + VIS	2 m	24	15	B8, B4, B6, B3, B1, GNDVI, MCARI, NDVIRE, B7, RVI, MTVI2, B5, CMRI, MSAVI, NDWI
5	WV-2 SR + TFs	2 m	36	18	Simple6, simple5, higher3, advanced5, B7, simple7, B6, simple4, B3, simple2, higher8, advanced10, B5, B4, higher7, B2, advanced1, higher5
6	WV-2 VIS + TFs	2 m	44	25	Simple6, advanced7, higher3, advanced6, higher8, advanced5, RVI, simple7, simple4, higher9, advanced10, GNDVI, NEVIRE, DVI, simple2, MCARI, advanced2, higher7, MTVI2, CMRI, advanced1, higher5, higher6, EVI2, NDWI
7	WV-2 SR + VIS + TFs	2 m	52	22	Simple6, simple5, advanced5, higher3, B6, higher8, simple2, simple4, B3, simple7, advanced9, advanced10, RVI, B4, GNDVI, higher7, B5, advanced1, higher5, CMRI, higher6, NDWI
8	OHS SR	10 m	32	19	B30, B1, B17, B32, B7, B16, B27, B2, B10, B4, B15, B23, B18, B13, B20, B14, B19, B21, B22
9	OHS FDR	10 m	32	23	Dif_B19, dif_B31, dif_B18, dif_B2, dif_B29, dif_B3, dif_B26, dif_B9, dif_B12, dif_B30, dif_B21, dif_B16, dif_B10, dif_B22, dif_B20, dif_B24, dif_B11, dif_B6, dif_B13, dif_B23, dif_B15, dif_B8, dif_B17
10	OHS VIS	10 m	23	9	MCARI2, MSR2, MCARI1, RVI, MSAVI, Clg2, GNDVI, RDVI, CMRI
11	OHS SR + FDR + VIS	10 m	87	28	Dif_B29, B32, dif_B19, dif_B32, dif_B12, B16, dif_B20, dif_B23, MSR2, B1, dif_B25, B2, dif_B2, dif_B9, dif_B13, dif_B22, dif_B21, dif_B3, B10, dif_B16, B15, B23, MCARI, B18, B11, dif_B15, B21, B22
12	ALOS-2 BI	6.38 m	5	5	HH, HV, HH-HV, HH+HV, HH/HV
13	ALOS-2 PFs	6.38 m	7	7	Entropy, alpha, $(1 - H)A$, $(1 - H)(1 - A)$, HA , $H(1 - A)$, anisotropy
14	ALOS-2 BI + PFs	6.38 m	12	8	HV, HH, entropy, alpha, HH + HV, HA , $H(1 - A)$, Anisotropy
Combined data					
15	WV-2 + OHS	2 m	139	34	Dif_B29, OHS_B16, OHS_B1, higher3, OHS_MSR2, dif_B22, dif_B16, WV2_B3, simple7, OHS_B32, OHS_B13, dif_B9, higher8, dif_B3, dif_B2, simple3, advanced9, simple4, advanced6, GNDVI, simple2, higher7, WV2_B5, WV2_B4, _WV2_B6, higher5, MCARI, NTVI2, advanced7, advanced8, RVI, WV2_B2, simple5, NDWI
16	WV-2 + ALOS-2	2 m	64	36	Entropy, simple6, HV, B3, HH, simple3, higher3, HH+HV, higher8, advanced6, B4, advanced5, simple4, B6, simple7, simple5, advanced10, advanced7, higher10, B2, B5, simple8, higher7, GNDVI, advanced2, simple2, simple1, advanced1, MCARI, higher5, B7, RVI, CMRI, NDVIRE, EVI2, higher6

(Continued)

Table 2. (Continued)

No.	Scheme	Resolution (m)	Raw feature size	Optimal feature size	Selected features (listing in descending order of importance)
17	OHS + ALOS-2	10 m	99	40	HV, $H(1 - A)$, HH, dif_B31, dif_B23, dif_B21, dif_B19, dif_B20, dif_B12, dif_B22, dif_B2, B2, dif_B29, B1, B16, dif_B3, dif_B11, dif_B8, dif_B10, dif_B25, dif_B9, B31, MSR2, dif_B1, B7, B17, dif_B16, dif_B15, B15, dif_B13, C1g1, MCARI, B14, B8, B10, B20, B13, B12, MCARI1, dif_B18
18	WV-2 + OHS + ALOS-2	2 m	151	62	Dif_B29, B3, dif_B22, simple6, OHS_B1, higher3, OHS_B17, OHS_B16, simple3, OHS_B32, dif_B6, OHS_MSR2, OHS_B2, dif_B20, dif_B9, dif_B2, HH+HV, dif_B23, dif_B3, higher8, B6, B5, OHS_B13, entropy, dif_B11, dif_B1, B4, GNDVI, dif_B15, alpha, simple7, dif_B26, advanced5, advanced6, higher7, dif_B16, OHS_B18, OHS_B12, HH, simple2, simple8, advanced9, advanced7, MTVI2, advanced10, advanced8, simple4, MCARI, B2, higher9, RVI, higher10, simple5, advanced2, NDVI, B1, advanced1, higher6, higher2, higher1, high4, NDWI

data were used to classify mangrove species, integrating SR and TFs of WV-2 data obtained higher accuracy (OA = 88.31%, QD = 9.63%, and AD = 2.06%) than the other schemes of the WV-2 (OA = 75.96% to 88.29%). TFs significantly improved the OA of WV-2 image from 75.96% to 82.29% to 88.31%. OHS produced lower accuracy (OA = 74.22% to 86.52%) than the WV-2 data. The ALOS-2 data generated the lowest accuracy with OA = 47.09% to 54.48%.

Integrating any 2 types of data from WV-2, OHS, or ALOS-2 generated a higher classification accuracy than using stand-alone data. The combination of WV-2 and OHS data achieved the highest accuracy, 5.2% (OA) higher than that of WV-2 alone and 6.53% (OA) than that of OHS alone, followed by the combination of WV-2 and ALOS-2; the combination of OHS and ALOS-2 obtained the lowest accuracy. The highest classification accuracy was achieved using the 3 data sources with OA = 94.02%, QD = 4.44%, and AD = 1.54%, which was 5.71% higher than WV-2 alone, 7.5% higher than OHS alone, and 39.54% higher than ALOS-2 alone.

From the results derived from WV-2 (Table S6), it can be observed that KO was difficult to classify correctly with PA of 0.31 to 0.41 and UA of 0.08 to 0.09 using SR features. This was improved with TFs, significantly increase in around 0.52 in PA and 0.55 in UA. TFs also had a positive effect on the discrimination of other species, especially for species with low accuracies such as AC and SA. The accuracies using the spectral features of the OHS data alone were higher than that of WV-2. In addition, ALOS-2 alone cannot effectively distinguish among mangrove species. In most cases, the combined data yield higher accuracies for species classification than the single-source data, among which the 3-source data achieved the highest accuracy.

Mapping the distribution of mangrove species

The mangrove species distribution map produced by scheme 18 was shown in Fig. 6. The study area covers a total of 537.29 hm^2 , of which 527.14 hm^2 is covered by mangroves dominated

by AC, AM, BG, and RS, covering 150.06, 135.50, 79.48, and 71.92 hm^2 , respectively. SA and KO were found smaller distributed over 41.86 and 3.93 hm^2 , respectively. AC is a pioneer species, with a high degree of population aggregation, and is usually distributed in different tidal zones in the central and southeastern parts of the study area. AM mainly occurs in the middle and low tidal zone in the north west of the study area, along the eastern coast of the north in the bank, and in the low and middle tidal zone along the central river. It is close to each other where BG and RS are distributed, mainly in the central northwestern and in the central and southeastern areas near the shore embankment. SA is mostly situated in the northeastern and southeastern banks of the study area. Nevertheless, KO is the least distributed and is often mixed with AC or BG.

In terms of data source, WV-2 had obvious advantages [Fig. 7 (1 to 7)], performing best using SR and TFs, whereas ALOS-2 data, in contrast, produced the poorest results [Fig. 7 (12 to 14)], whose pepper effect was particularly pronounced. KO was indiscernible using WV-2 data without TFs [Fig. 7 (1, 2, and 4)] but could be identified by adding TFs or utilizing OHS data. The same scenario occurred with AC and SA. We also found that RS was overestimated using only the TFs of WV-2 at the intersection of BG and RS, as shown in Fig. 7 (3). For species with similar canopy heights but different canopy structures (such as AC and AM) distinguishing between them can be done using TFs and more detailed spectral and FDR information of OHS, as shown in Fig. 8 (1 to 11).

Discussion

Contributions of multisource remote sensing data

Our results demonstrated that combining features retrieved from 3-source remote sensing data yielded the highest classification results, with an OA of 94.02% compared to the WV-2 (88.31%), OHS (86.52%), and ALOS-2 (54.48%) features alone. This indicated that each remote sensing data had a positive

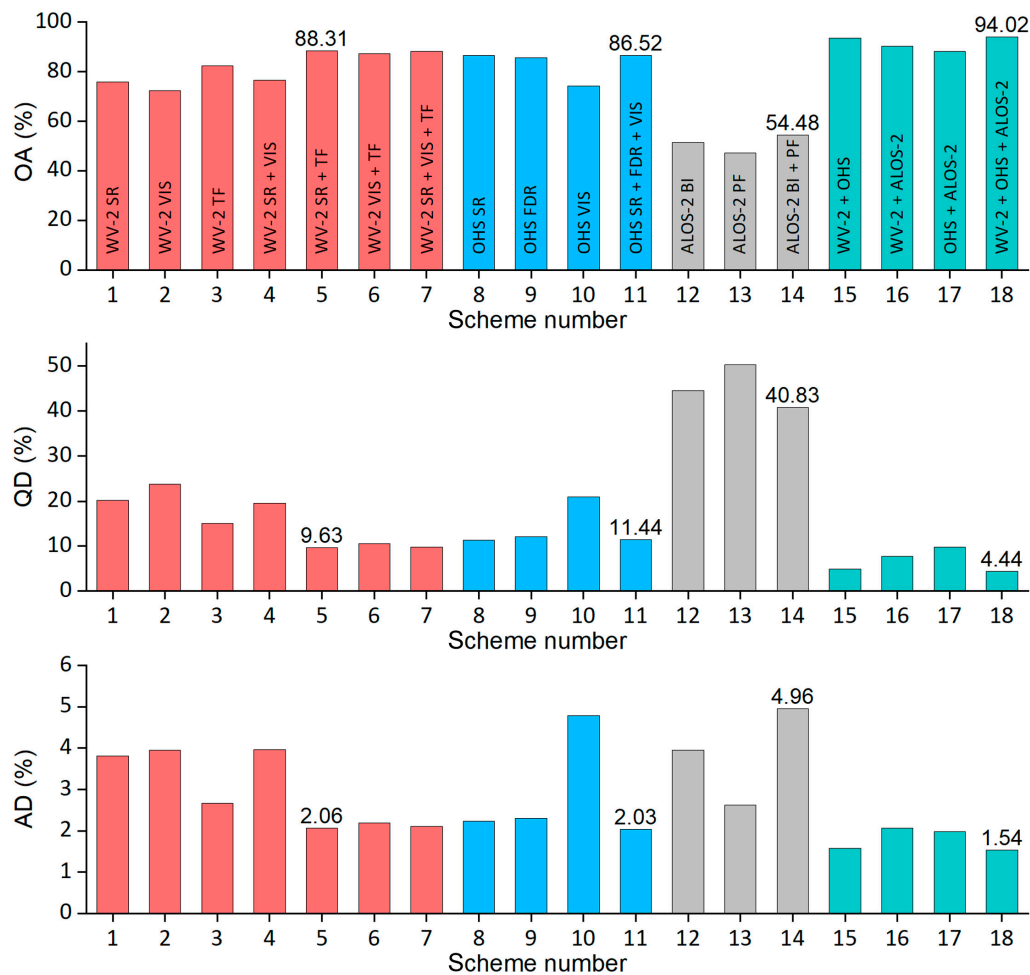


Fig. 5. Overall classification results using 18 feature sets.

contribution to mangrove species discrimination, which is consistent with previous studies [14,34].

In addition to the bands generally available in high-resolution images, WV-2 adds 3 special bands, in which the yellow and red-edge bands are more sensitive to chlorophyll and contribute to mangrove species classification [12,24]. Therefore, the ultra-high submeter spatial resolution and rich spectral bands allow for more accurate classification results. In this study, SR, VIS, and TFs from WV-2 image were generated for better mangrove species discrimination (see Tables S3 and S4). Each of them provided important supplementary information, and that especially TFs was found to outperform both SR and VIS, which describe the horizontal structure of images and reveal spatial variations, such as the compactness and hierarchy of mangrove species [8,35]. As a result, the 3 WV-2 features improved the OA and reduced the QD and AD by 0.81% to 15.88%, 0.74% to 14.05%, and 0.08% to 1.84% (Fig. 5, i.e., the comparison among schemes 1 to 4, 6, and 7). PA and UA for KO, AC, and SA also showed remarkable enhancements (Table S6). These results are consistent with those of previous studies [11]. It is worth noting that the VIS did not have a marked impact, and even the accuracy using only SR and TFs was already comparable to the accuracy using all WV-2 features. This may be because we utilized higher and advanced TFs, which better measure attributes such as texture clarity, roughness, and uniformity of clarity, whereas

VIS tend to saturate across mangrove species, resulting in little distinguishability.

Our results showed that OHS did not perform as well as WV-2 because the 10-m spatial resolution was not fine enough to account for the heterogeneity of mixed mangrove composition and field survey plots could not match very closely with the pixels on imagery [36]. However, both the combination of OHS and WV-2 and the combination of OHS and ALOS-2 data significantly improved the classification accuracy compared to the single data (Fig. 5). In terms of feature variables, both FDR and spectral bands of OHS showed preponderance, mainly because more spectral details and more features were generated. In particular, spectral derivative techniques (FDR used in this study) have been confirmed to differentiate overlapping signatures and reveal the spectral peak characteristics [14]. Differentiating between BG and RS can sometimes be challenging, as they share the similar size and structure of the folium. The addition of the OHS information improved accuracy of identification (Table S6). As shown in Fig. 3A and B, there are slight differences in visible SR in the WV-2 image among the 6 mangrove species, while more distinguishing features could be generated in the OHS image for model training to improve the species classification accuracy.

The complementary role (ALOS-2) of optical data for the improvement of mangrove species classification was assessed

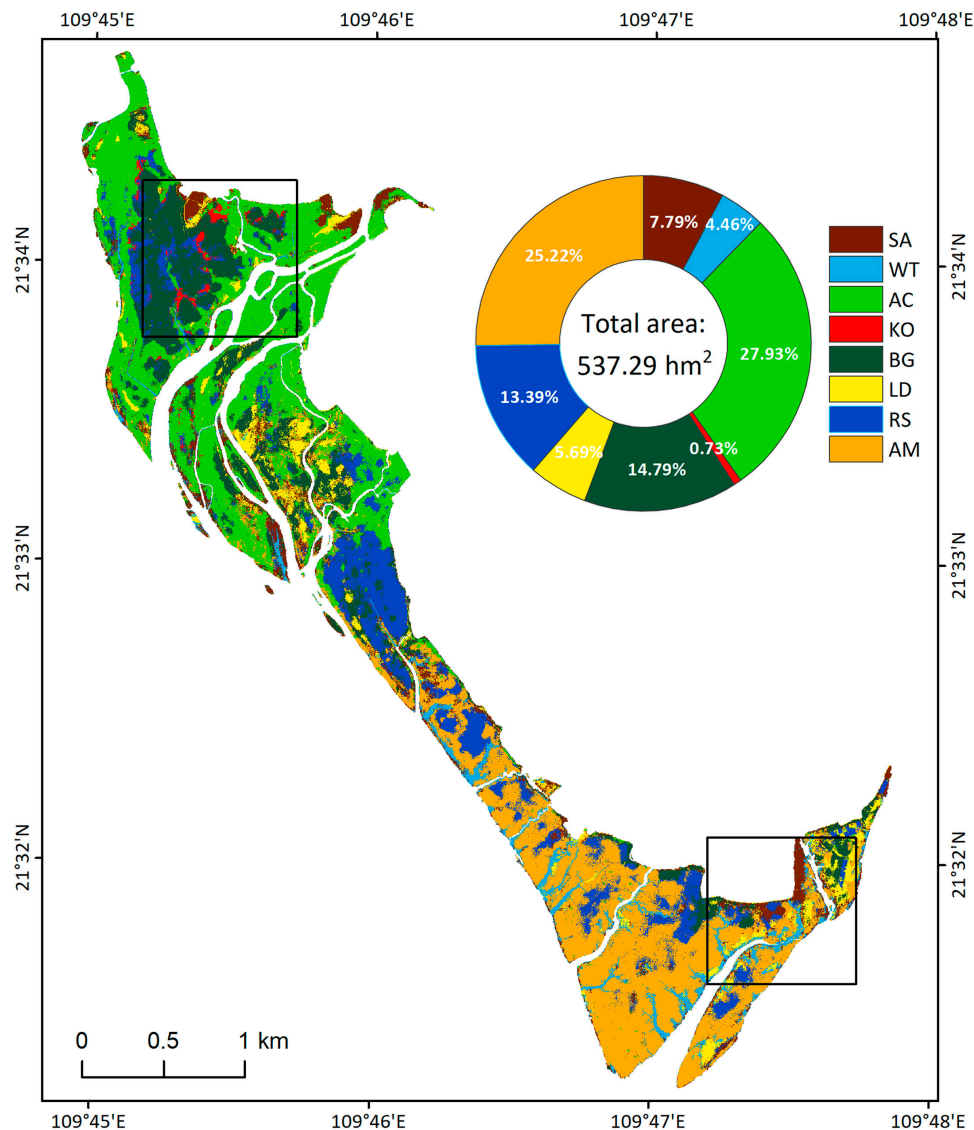


Fig. 6. The classified map of mangrove species distribution produced by scheme18. The black rectangles in the top left and bottom right are the positions of the classification details in Figs. 7 and 8, respectively.

in this study. OA was improved from 88.13% to 90.25% (schemes 7 and 16 in Fig. 5) and from 86.52% to 88.29% (schemes 11 and 17), QD was reduced by 2.07% and 1.71%, and AD was reduced by 0.05% and 0.05%, respectively. Although the improvement may not be substantial (approximately 2%), it may be imperative to effectively map and monitor mangroves on a large scale [31]. This improvement can be further interpreted from the PA and UA among different species, illustrating that the double polarimetric SAR data mainly contributed to the discrimination of KO, AC, and AM species. In general, the spectral similarity among mangrove species is high, which poses a challenge for species classification using optical data alone [13]. However, they are structurally diverse, and SAR data (e.g., the L-band in this study) can penetrate mangrove canopies and provide morphological information (e.g., tree height, canopy closure, shape, and size of leaves) by utilizing surface, double bounce, and volume scattering mechanisms [7]. Among the 6 species, mature KO is often the tallest, crowns are often small and loose in shape, and leaves are usually deep green with

a high chlorophyll content [24]. AC is usually dwarf and appears similar to AM in terms of leaf size and shape of their leaves. This would explain how SAR data compensate for the classification of different mangrove species.

With the above analysis, owing to spectral similarity and structural complexity, single-source remote sensing data have its limitations for mangrove species classification. Our study showed the notable potential of combining ultrahigh-resolution multispectral, hyperspectral, and L-band SAR data in mangrove species classification, which can overcome the shortage of low discriminability of feature variables derived from single-source data.

Evaluation of XGBoost classifier

Several studies have explored the use of the XGBoost algorithm in remote sensing application of mangrove; however, the study object of the current study was the inversion of physical and chemical parameters [37] and very few on mangrove species classification using single type, source, or UAV remote sensing

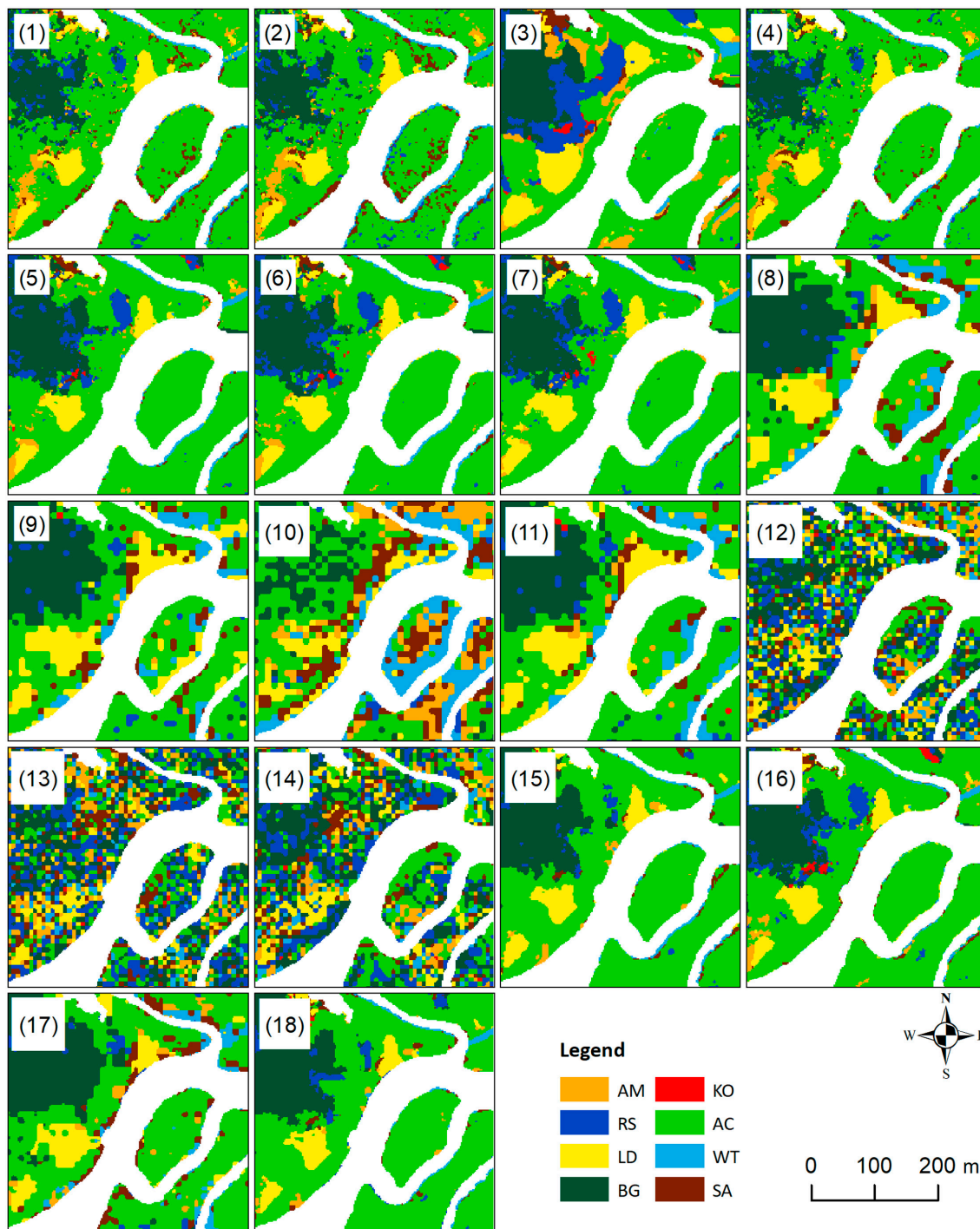


Fig. 7. A comparison of mangrove species distribution maps derived from different data combinations (schemes 1 to 18).

data. On the basis of our knowledge, this is the first study on the use of the XGBoost algorithm and 3-source spaceborne data for mangrove species classification. One of the most critical issues of ML methods is unstable model results [21]. Given the model reliability, we performed 100 rounds of random experiments during the feature selection and accuracy evaluation according to the random grouping of training and test samples. Currently, there is no clear evidence for determining the best approach for mangrove species classification [7]. This is due to the fact that there are many factors that can influence

the classification accuracy, and the classification results are often poorly to compare across different sites, species, data, and algorithms used. Historically, SVM and RF were the most widely used algorithms for mangrove species classification [9,13]. However, their application to images with numerous features, such as hyperspectral images or training samples with high noise, is limited. Apart from high accuracy and efficient computation, XGBoost can handle imbalanced datasets, providing an advantage over RF [23]. Our results demonstrated that the XGBoost classifier successfully classified the

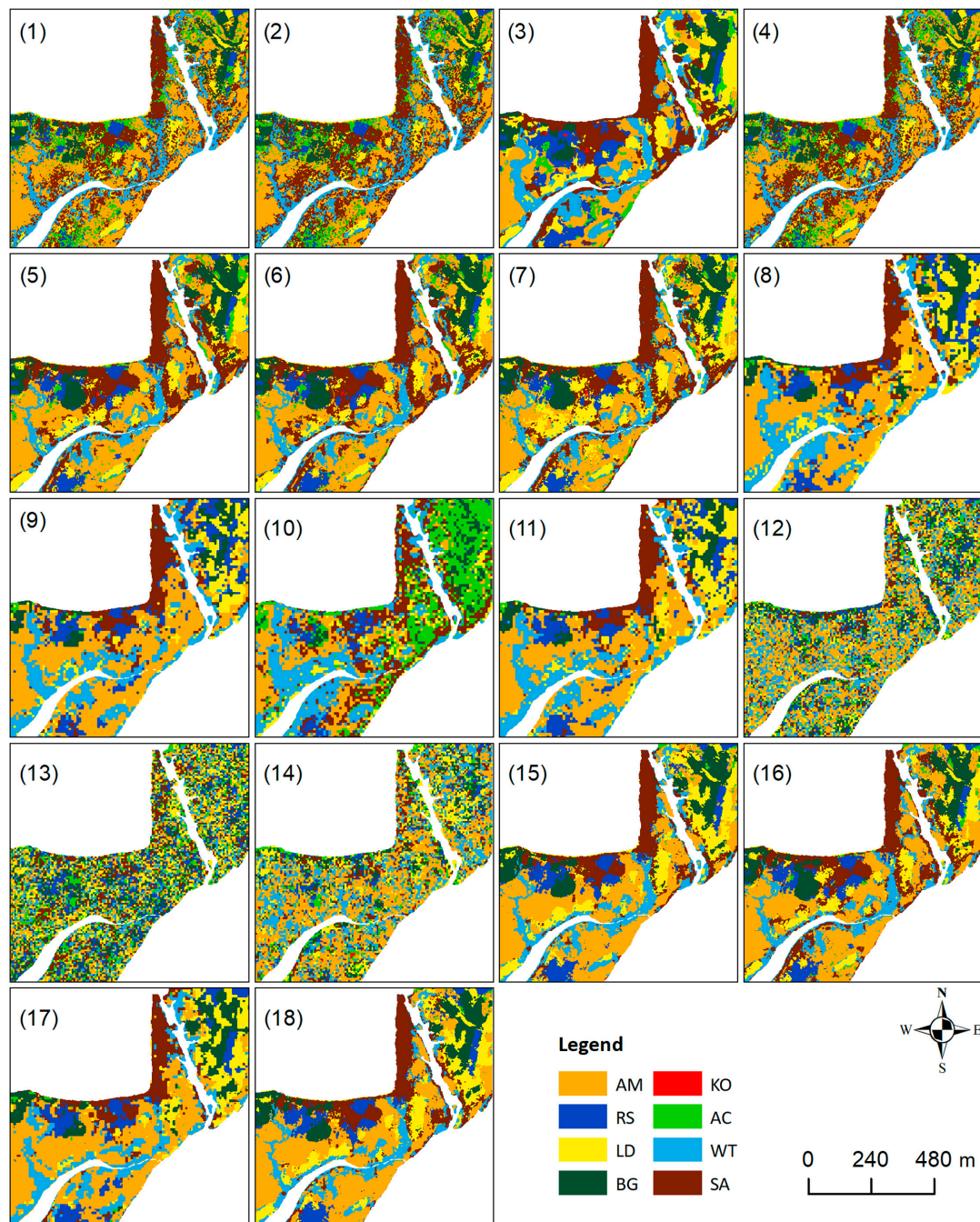


Fig. 8. A comparison of mangrove species distribution maps derived from different data combinations (schemes 1 to 18).

6 mangrove species with a high accuracy of 94.02% using a combination of 3-source data. Cao et al. [14] incorporated UAV-based hyperspectral and light detection and ranging data to classify mangrove species using the rotation forest ML algorithm, achieving the highest OA of 97.22%. However, it is worth noting that their study site was only 2.65 hm², and the identified mangrove species were mostly artificially planted. It can be further confirmed that XGBoost has powerful potential for mangrove species discrimination based on visual classification performance (Figs. 6 to 8). Nonetheless, this assertion should be validated in future studies. In addition, the framework we proposed provides a way to apply the

high-dimensional features of remote sensing big data and mangrove species classification on a large scale.

Uncertainties and future research improvement

Our study succeeded in achieving high-precision classification of mangrove species by combining 3-source data. On account of the species diversity and spatial heterogeneity, multisource remote sensing data are undoubtedly an effective way to solve this problem [38]. There are still several potential limitations and uncertainties in our study.

Remote sensing data types and classification algorithms are the 2 most important factors for the accurate extraction of

mangrove species. Benefiting from higher spatial resolution, broad spectrum range, and rich texture information, most studies have been conducted based on WV instruments, which greatly improved the species classification of mangroves and other trees [11,12]. For mangrove forests in such special habitats, it is challenging to obtain clear optical satellite remote sensing data, particularly multisource satellite data at the same time. In addition to the collection time, the mismatch of spatial resolution among different data sources is a well-known challenge in data fusion or combination [20]. The 3-source remote sensing data utilized in our study were acquired at different temporal and spatial resolution. Although we conducted high-accuracy image registration with the WV-2 as a reference imagery, inevitably, the samples would be displaced in the hyperspectral and ALOS-2 data, and those were in correct locations in the WV-2 data. In addition, because of the field experimental conditions, there were inconsistencies between our field data and satellite images. These mismatches directly affected the accuracy of the classification model.

Our study showed the great potential of the XGBoost classifier for mangrove species identification, which can deal with a large number of features generated from multisource remote sensing data and achieve exceptional performance. The main reason for this is the scalability of the XGBoost algorithm across all situations, coupled with its fast-processing speed. The XGBoost-RFE approach was utilized in this study for the first time to identify sensitive variables for the classification of mangroves species, which not only reduces the redundancy of feature variables but also enhances the interpretability of the model. Interpretability is a key issue in ML, as higher levels of precision often lead to greater model complexity and decreased interpretability. In addition, it is often challenging to quantitatively assess the impact of various feature variables on the prediction outcomes of ML models. Except for the feature importance score operations provided by ML algorithms, some researchers, such as Kavzoglu and Teke [21], used the SHapley Additive exPlanations (SHAP) strategy to further improve the model interpretation, which not only identifies whether variables make negative or positive contributions but also calculates for each sample. This should be the focus of future studies.

The quality and quantity of samples used in the model are decisive in the classification and accuracy assessment of mangrove species. In this study, samples were derived from field survey and manual interpretation of high-resolution images. On the one hand, it is difficult to collect samples at mangrove species level and more difficult to achieve a balanced sample across species. For example, we collected 16 filed samples of KO due to their small and fragmented distribution. On the other hand, manually interpreted samples can be highly dependent on the influence of the professionalism and subjectivity of the interpreters, which may affect the objectivity of sample quality. One way to address this challenge is to augment the samples through transfer learning techniques [39], which will be attempted in the future.

In addition, several studies have revealed that even slight differences in elevation can have a substantial impact on the distribution and growth of mangrove species [40,41]. In the present study, obvious differences in elevation among the 6 mangrove species are shown in Fig. S37. The classification accuracy of mangrove species may be further improved by using plant functional traits such as equivalent water thickness [42] and leaf chlorophyll [43]. Therefore, it is necessary to explore the impact of incorporating these features on mangrove species

discrimination in future work, as it can provide an in-depth understanding of the classification model.

Conclusion

Mapping mangrove species accurately is crucial to mangroves restoration and sustainable development. In this study, we developed a novel framework for mangrove species classification in ZMNNR, China, by combining the XGBoost classifier and the XGBoost-RFE algorithm based on 3-source satellite data. We found that combined data (any 2 or 3 data sources) can effectively identify the 6 mangrove species and generate higher classification accuracy than single data. The highest accuracy was achieved by utilizing 3 data source features with OA = 94.02%, QD = 4.44%, and AD = 1.54%. Results of feature selection for the 18 schemes indicated that texture information generated by WV-2 was the most important variable for mangrove species classification. In addition, the FDR and spectral bands from OHS, as well as the backscatter coefficients of ALOS-2, were especially conducive to mangrove species discrimination, as they provided important supplementary information for each other. The XGBoost algorithm, with both feature selection and classification functions, has a great potential for mangrove species identification.

Classifying mangrove species is a challenging task due to the similarities in spectral properties and complexities in the structural features of different species. Our findings provide crucial data to support the conservation and restoration of mangrove ecosystems and hold important scientific significance and operational value. In future research, we aim to incorporate elevation and plant functional traits into the classification model, while also directing our focus towards sample migration and model interpretability.

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Data Availability

The manuscript used to select the window size of TFs, select sensitive features for discriminating mangrove species, and

classify mangrove species are publicly available at <https://github.com/zxp-dpz1122/JRS01>.

Supplementary Materials

Figs. S1 to S37
Tables S1 to S6

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Performance of XGBoost Ensemble Learning Algorithm for Mangrove Species Classification with Multisource Spaceborne Remote Sensing Data

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Accurately and timely monitoring the spatial distribution and composition of mangrove species is critical for assessing mangroves' health, dynamics, and biodiversity, as well as mangroves' degradation and restoration. Recent advances in machine learning algorithms, coupled with spaceborne remote sensing technique, offer an unprecedented opportunity to map mangroves at species level with high resolution over large extents. However, a single data source or data type is insufficient to capture the complex features of mangrove species and cannot satisfy the need for fine species classification. Moreover, identifying and selecting effective features derived from integrated multisource data are essential for integrating high-dimensional features for mangrove species discrimination. In this study, we developed a novel framework for mangrove species classification using spectral, texture, and polarization information derived from 3-source spaceborne imagery: WorldView-2 (WV-2), OrbitalHyperSpectral (OHS), and Advanced Land Observing Satellite-2 (ALOS-2). A total of 151 remote sensing features were first extracted, and 18 schemes were designed. Then, a wrapper method by combining extreme gradient boosting with recursive feature elimination (XGBoost-RFE) was conducted to select the sensitive variables and determine the optimal subset size of all features. Finally, an ensemble learning algorithm of XGBoost was applied to classify 6 mangrove species in the Zhanjiang Mangrove National Nature Reserve, China. Our results showed that combining multispectral, hyperspectral, and L-band synthetic aperture radar features yielded the best mangrove species classification results, with an overall accuracy of 94.02%, a quantity disagreement of 4.44%, and an allocation disagreement of 1.54%. In addition, this study demonstrated important application potential of the XGBoost classifier. The proposed framework could provide fine-scale data and conduce to mangroves' conservation and restoration.

Image

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